

Functional Safety and Industrial Communication Associated for the Smart Factory

T.K. Tran, H. Yahoui, and N. Siauve

University of Lyon
AMPERE CNRS 5005, University Lyon1
69622 Villeurbanne, France

e-mail: thanh-kha.tran@univ-lyon1.fr / hamed.yahoui@univ-lyon1.fr / nicolas.siauve@univ-lyon1.fr

Abstract—Traditionally, machines safety were realized with hardwired circuits; today, there is a growing trend to use programmable electronic technology such as programmable logic controllers (PLCs). The present paper introduces an innovative platform designed to be used in industrial automation practical work. The platform includes the different technologies used in functional safety: safe relays, safe programmable logic controllers that can be programmed with the specific software of each manufacturer, various safe sensors, and actuators. For building a safety application system with a safety PLC or a safety relay and eventually an associated safety network, users shall consider to not only a safety programming but also technology, configuration, installation, operation, and maintenance. So the platform allows to students to acquire the theoretical knowledge and experimental technical skills in industrial safety and to realize the choice, design and the setting of the appropriate structure of the control system for the required availability and safety. The proposed laboratory experimental pieces of training are focusing on functional safety in factory automation with the last technologies used in the industry. The various experimental stations of the platform are described. This innovative platform developed will be employed by Bachelor and Master students. Students are part-time workers for one or two years in the industrial engineering domain at the Electrical Engineering department of University Claude Bernard of Lyon, France.

Keywords—*continuing education, control engineering education, industrial programmable logic controllers (PLCs), electrical engineering education, industrial training, functional safety.*

I. INTRODUCTION

The safety could be defined as the absence of risks to plant employees or the public. In the absolute, safety would be easily achieved by the entire suppression of risk. Safety is achieved by the decrease of risk to an acceptable or tolerable level. This paper will only deal with the safety of machines used in industry (machines and lines in the factories, packaging machines manufacturing process, machine tools, standard specialty machines, plastics processing machines, ...).

The method used in industries to decrease the risk of the machine in factories includes the following two aspects. The first one is the intrinsic safety in which the workers and

machines shall be separated physically by a guard, a locked door, specialized insulation, or other protective physical devices. In this first case, the system is protected physically from dysfunction. This point corresponds to passive measures. Moreover, the second is the functional safety, for this second point the safety control system shall stop machines in the hazardous area when it detects anyone intruding into a hazardous area or dysfunction of the system [1]. About this second point, it corresponds to active measures. Mostly the safety is a combination of these two measures. The active measures constitute the functional safety that is a concept applicable across all industry sectors (petrochemical, pharmaceutical production, power generation, chemical, plastic, food, ...) [2]. Functional safety is fundamental to the enabling of sophisticated technology used for safety-related systems. It provides the certitude that the safety-related systems will offer the necessary risk reduction required to achieve safety for the machine. So it is achieving safety for the machine giving rise to the hazards.

The functional safety is a part of the overall safety that hinges on the correct running of the equipment, machine, or process in according to its inputs. Rather than a tedious definition, a short example allows clarifying merely the meaning of functional safety. We can take the case of a system which consists of a tank which is filled without the controller. The intrinsic safety is to include in the design of the system a permanent physical device (under the form of a drain pipe) to ensure the discharge of the liquid in the event of accidental over-filling of the tank, to avoid its overflow while a solution incorporating level sensors, solenoid valves, and control module (with specifications that meet the safety standards) that will empty the tank to avoid its overflow will be considered as a functional safety solution.

The functional safety system needs an association of safety input devices (sensor detecting), safety circuits (logic evaluating) and safety output devices (actuators, alerting). All the safety devices are interconnected each other via a communication bus or a hardwired link to form a safety system, which performs a specific safety-related function (Fig. 1). This is functional safety. Therefore the functional safety of

an electrical control system is highly relevant to the control of hazards arising from moving parts of the mechanism.

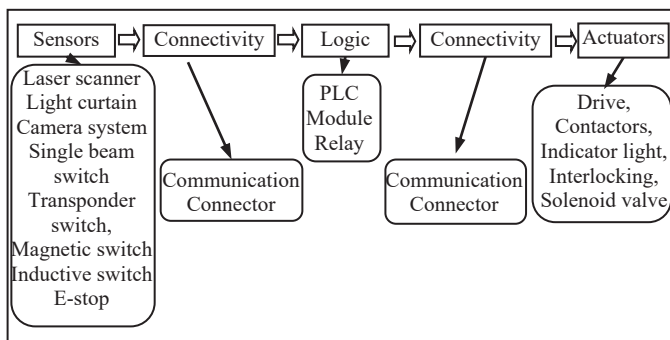


Fig. 1. The framework of functional safety with the different categories of safe devices.

The safety function and safety integrity are the two requirements that are necessary to achieve functional safety. Risk assessment has a crucial role in developing functional safety requirements. Hazard and task analysis give the functional requirements for safety (i.e., the safety function). Moreover, the risk quantification leads the safety integrity requirements (i.e., the safety integrity or performance level).

This paper aims to introduce a new experimental platform used to train students on functional safety. The automation safety platform has been designed and developed to take into account the problems of the factory automation area (including the machine manufacturers). The other activity areas, such as building automation, infrastructure automation, or process control are not considered. The objective of this platform is to answer the needs of industries by training bachelor and master students for their future jobs. Functional safety is required in a continuously increasing amount of industrial applications. The functional safety area is a growing field. Students must be trained on these technologies and must acquire the associated skills to develop their employability. The manufacturers meet some difficulties to find young graduate students with safety skills.

To assess the level of knowledge and competence of students in the field of functional safety, a questionnaire was conducted. The survey was conducted during the school year in September 2015 at the University Lyon 1. Third-year bachelor students complete the questionnaire in initial training (47 students from 11 different training or educational establishment for their second year of bachelor) and professional training: sandwich courses (27 students coming from 15 different training or educational establishment for their second year of bachelor). All students have a curriculum in the field of automatic, electrical, and electronics.

The questionnaire consists of five questions: - Q1. During your academic courses had you follow safety training, - during your company internships: Q2. Had you seen safety devices, Q3. Had you used safety devices, Q4. Had you developed a safety application, Q5 Had you written up a safety note. Figure 2 presents the synthesis results of this questionnaire

survey for the students in professional training (27 students). Concerning the students in initial training among the 47 students, only one has replied positively to the questionnaire at the question Q3 (used safety devices), the considered student replaced a programmable safety relay during a maintenance operation.

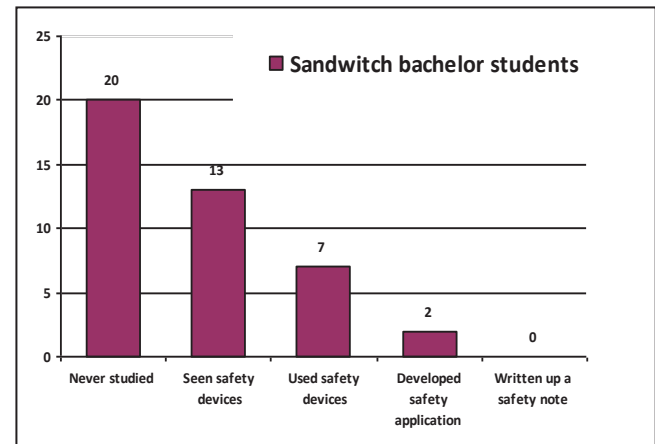


Fig. 2. Questionnaire synthesis.

The results of the study showed a weakness of student knowledge in the field of functional safety by their previous training. It was therefore decided to develop this teaching by the establishment of a theoretical course and development of a series of possible work manipulations. The developed labs will be used in a new bachelor training entitled Intelligent Interconnected Controlled Systems (in French S2IP Systèmes Intelligents Interconnectés et Pilotés) which will begin in September 2016. The teaching on safety with the different pieces of training developed will begin in September 2017.

The students will be part-time workers in bachelor or eventually in master degrees, in the industrial engineering domain. The platform includes seven labs (six experimental and one theoretical) with different levels according to the students are in bachelor and master degrees. The duration of each lab is 4 hours. The six experimental labs cover all the safety control technological solutions, from the historical hardwired solutions to new prototype devices, used in industrial automation: safe PLCs, safety relays, safe relays programmable, safe modules and this for several constructors. The different specific safety software and safety communication protocols associated with each constructor will be studied and used. The different inputs and outputs safety devices of the safety controller such as: sensors (transponder switch, magnetic switch, inductive switch, laser scanner, light curtain, camera system, single beam switch, non-contact door monitoring) and actuators (safe drives, safe contactors, door monitoring with high locking force interlocks) are also part of the teaching.

With these different pieces of training, the students acquired the knowledge and competencies to conceive safe systems apply to factory automation integrating efficiency, flexibility,

availability, reliability, scalability, durability, and performance requirements. The students will be able to design and develop safe applications that ensured the complete safety of the plant employees, allowed to drives and actuators to perform intelligent and safe reaction that provide controlled emergency operation with rapid restarts and obtained the shortest response times that guaranteed minimum stopping distance, even for very fast movements [3].

This paper is structured as follow. Section II presents an introduction of some fundamental aspects of functional safety for machines and the technical requirements. Part III deals the different strategies used for the functional safety for the control system (PLC or relay solutions). The parts II and III represented the outline of the content developed during the technical courses. Section IV describes the experimental setup of practical works designed and developed and also the associated skill. Finally, the conclusions and perspectives will be exposed in section V.

II. SAFETY STANDARDS

Since 2000, the works on safety have been intensified within ISO. In 2009, appear news standards EN ISO 13849-1 and EN 62061 to determine the PL (Performance Level) or SIL (Safety Integrity Level) respectively. Since 2010 safety components manufacturers offers news solutions each year. The machines manufacturers must protect machines and persons against accidents following legal requirements for machines, safety-relevant directives, regulations, and standards, selection and application of safety devices. This section deals with safety standards and safety analysis methods.

A. Safety Legislation and Standards

December 31, 2011, came into force the new Machinery Directive (2006/42 / EC) that regulates the assembly and trade of machinery in the EU. To meet the requirements of this regulation, it is necessary to comply with the harmonized standards EN / IEC 62061 or EN ISO 13849-1. Now, this new Directive takes into account all types of technologies, mechanical, hydraulic, pneumatic, electro-pneumatic, and electric. The machine manufacturer must choose between these two standards depending on its application. The IEC 62061 standard replaces the 954-1 with the categories (B, 1, 2, 3 and 4) which indicated the required behavior of an electrical control circuit in the event of a fault. Now designers can comply with EN ISO 13849-1 or EN 62061 to demonstrate the conformity of their machines with the machinery directive 2006/42 / EC. These two new standards take account of the possible occurrence of a failure, but also of its probability. The EN ISO 13849-1 standard applies to all technologies (hydraulic, pneumatic, mechanical, ...). The EN ISO 13849-1 standard relies on the well-known architectures (categories B, 1, 2, 3, 4) of the old EN 954-1 standard and the Safety Analysis Method to determine the level of performance achieved: PL_a expresses the Performance level to PL_e. However, the EN 62061 standard applies only to safety-related electrical, electronic, and programmable electronic control

systems. Performance capability is designed by the Safety Integrity Level (SIL) which ranges from SIL1-4 to SIL 4.

B. Safety Analysis Methods

The risk management is an iterative method that involves three major phases: the analysis phase, the design phase, and the finalization phase (fig. 3). Steps one and two are carried out iteratively until the desired level of security is achieved.

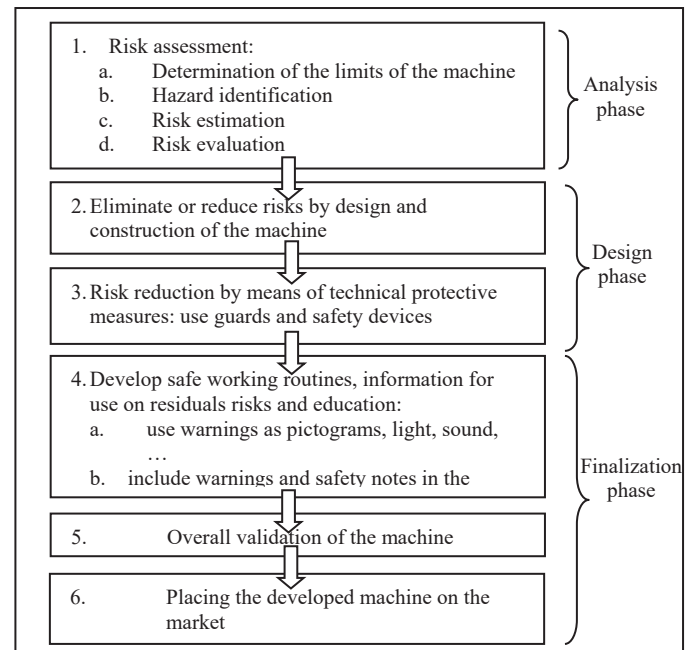


Fig. 3. Risk management.

III. SAFETY STRATEGIES OPTIONS

To ensure the functional safety of a system, different configurations of automated control systems can be employed: - wire safety based on relays components, - programmable safety module with optionally an additional industrial communications with the PLC (Ethernet or RS232), - integrated safety module or safe PLC. For all safety strategy options, the safety inputs/outputs are connected to safety-relevant sensors and actuators. Choosing the right option allows to reach the requested level of safety and can result in quicker time-to-market with higher product throughput, and lower total cost of ownership for safety systems, improving both overall equipment effectiveness, and return on assets for the manufacturers.

A. Dedicated Safety Relays

The safety relay is the simplest solution for integrating safety in an application. Typically two redundant contacts are wired to a safety relay (fig. 4) and eventually a separate ordinarily open contact is wired to the PLC for feedback indication (fig. 5). An encoder can be added with a specific card (sdm) in order to detect speed and direction of rotation or stop.

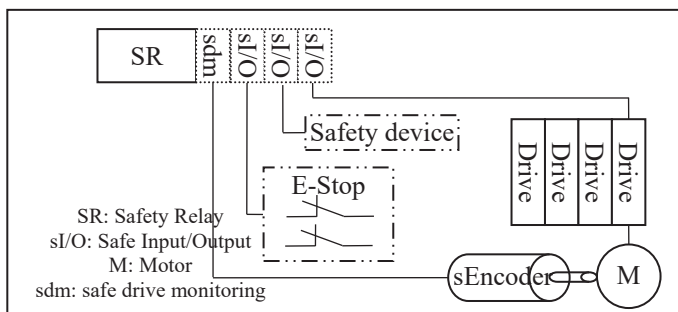


Fig. 4. The structure of the control system with hardwired (relay) safety solutions.

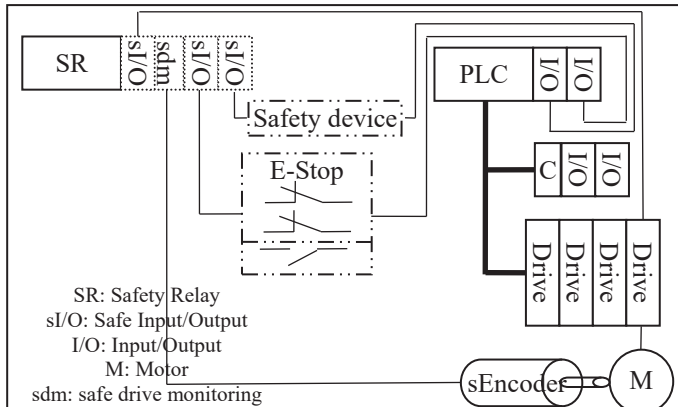


Fig. 5. The structure of the control system with hardwired safety and classical PLC.

On the other hand, when a machine required a high level of flexibility imposed by the presence of many possible machine options or when the system is composed of cascaded production machines a hard-wired reaches its limits without drastically increasing complexity or compromising safety. In this case, another technology must be envisaged by the manufacturers.

B. Software Configurable and Networked Safety Relays

The first option after the dedicated relays is the configurable safety relays solution. The configuration of the configurable safety relays can be realized historically from the screwdriver and now from the software. This solution can also be connected to a PLC from a communication bus (principally Ethernet), and the different safety devices (e.g., safety device 2 and 3) can also be connected each other by a specific interface bus (fig. 6).

C. Programmable Integrated Safety

The second programmable option is integrated safety. With this solution, the safety module is directly integrated into the PLC architecture. The highest level of machine safety can be reached with integrated safety. A significant advantage of an integrated safety solution is the suppression of the redundant cabling and the elimination of communication and external monitoring modules for safety solutions. The safety data are transferred on virtual wiring via the machine bus system in an intrinsically safe protocol (fig. 7).

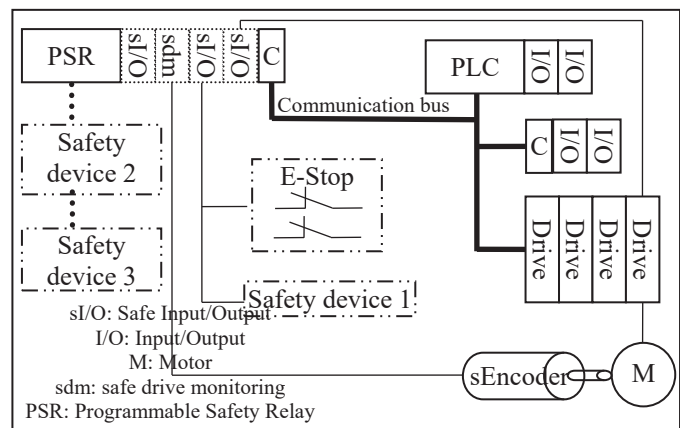


Fig. 6. The structure of the control system with programmable safety relay, standard PLC, an Ethernet communication bus between PSR and PLC, and a specific communication bus between PSR and safety device.

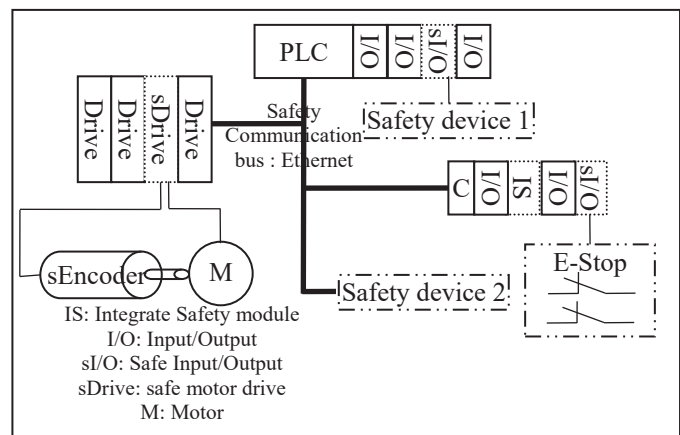


Fig. 7. The structure of the control system with integrates safety and an associated PLC.

From now on, even in the smallest application, programmable integrated technology is no more expensive than an ideal solution based on dedicated relay technology.

D. Dedicated Safety PLC

With this solution, a dedicated PLC can be used to fulfill the safety functions. In this case, several configurations are possible; figures 8, 9, and 11 presents the different configurations. Figure 8 presents the configuration with a specific PLC to control the safety, and the safety I/O are deported. The solutions on figure 9 integrate the safety I/O directly on the safety PLC that protect against burn out of the classical PLC or the bus controller (noted C on figure 8, 9 and 11). Moreover, with the last possibility exposed on figure 11, the classical PLC that controls the machining process is replaced by the safety PLC that ensure the control of the process and the safety of the system in the same device jointly.

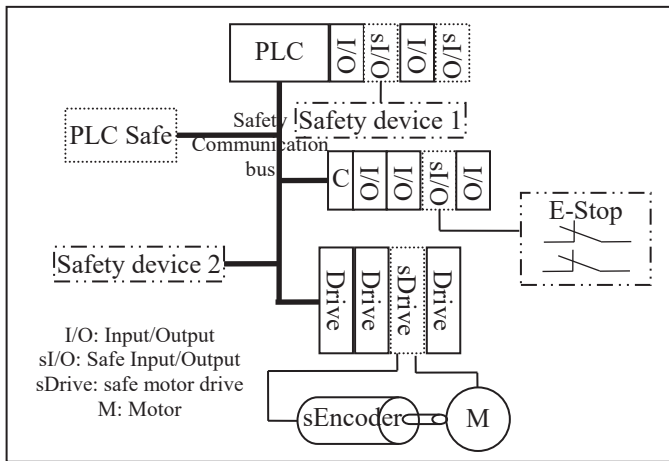


Fig. 8. The structure of the control system with specific safety PLC and a classical PLC.

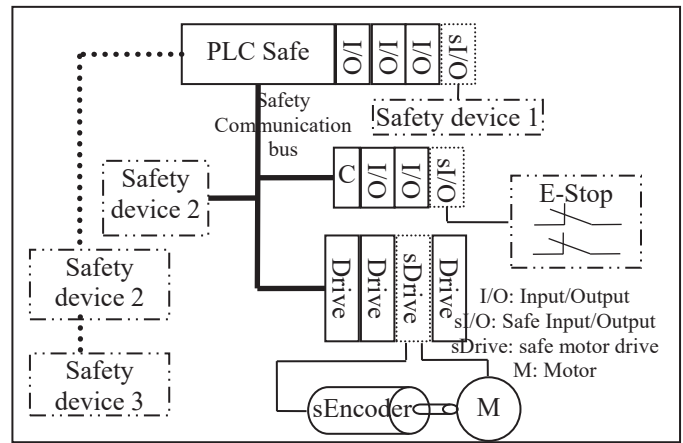


Fig. 11. The structure of the control system with specific safety PLC, which also includes the control of the system performed before by the conventional PLC in figure 8 and 9.

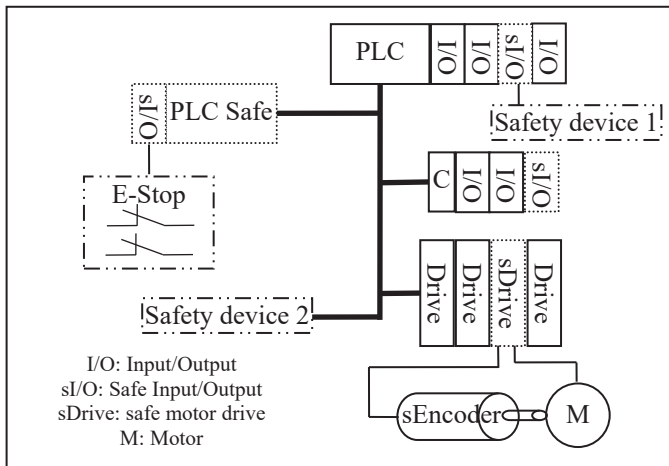


Fig. 9. The structure of the control system with specific safety PLC burn out and a classical PLC.

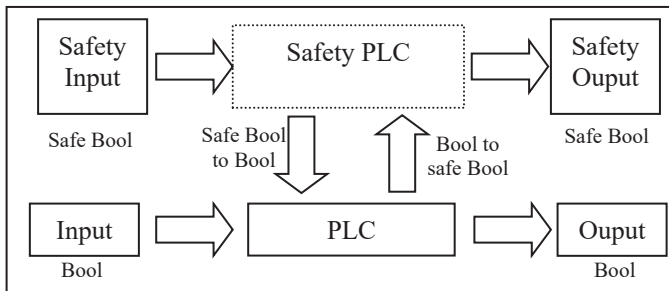


Fig. 10. Organization and exchanges between the safety program and the program managing automation.

Programmable safety technology based on semiconductor components is not subject to wear like dedicated hardwired safety technology that used electromechanical relays. Electromechanical relays must be replaced periodically to maintain the functionality of the safety application. Moreover, semi-conductors based safety technology is also considerably faster than relay based solution.

Figure 10 presents the specific structuring of the PLC program and integration of safe variables as well as the exchanges between the programs (safety program and classical programs) within the safety PLC. The safety management program is realized in functional block diagram language with a limited number of variable types and function blocks in order to limit the causes of errors by the designer.

IV. DEVELOPED PRACTICAL WORK SYSTEMS

The purpose of the different developed practical pieces of training is to enable to student to select the appropriate technology of safety control according to the application and know how to set up the safety control device and program the retained equipment in order to achieve the desired operation in compliance with the requirements of the standard.

The solid pieces of training developed are the results of several training courses with manufacturers and visits to industrial fairs (SPS IPC drives, Automatica, Motek and electronica) in order to target the relevant skills and associated equipment.

Seven experimental pieces of training on hardware (six pieces of training) and software (one training) are developed. The developed manipulations reproduce the different configurations described in section 3. The list of manipulations is as follows:

- Training 1: Use of the Sistema software to evaluate the PL performance level from a predefined user-defined architecture.
- Training 2: Wiring hardware Safety Relays (SR) and Programmable Safety Relays (PSR) corresponding to configurations of the figures 4 to 6 (fig. 12-13).
- Training 3: Programming a safety solution in a configuration that includes classical PLC and a specific interface communication between PSR and safety sensors associated with the solutions presented in figure 6 (fig. 16).
- Training 4-5: Programming a safety solution with an integrated safety module as exposed in figure 7 (fig. 14-15) for different PLC manufacturers.

- Training 6-7: Programming a safety solution corresponding to the configurations presented in figures 8, 9, and 11 with the modern software associated.
The figures 12 to 16 shows some pictures of developed manipulations.

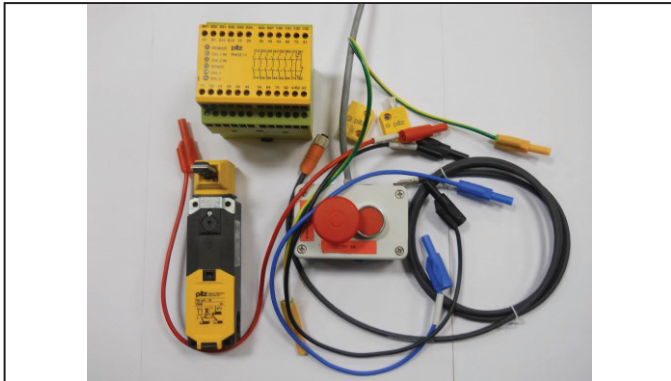


Fig. 12. Wiring a Safety Relay (SR) with safety actuators and sensors.



Fig. 13. Wiring of a Programmable Safety Relay (PSR) with safety actuators and sensors associated with a PLC.



Fig. 14. Programming a safety configuration which includes an integrated safety module and associated safety devices such as Light curtains, emergency stop, motor drive with enable input.

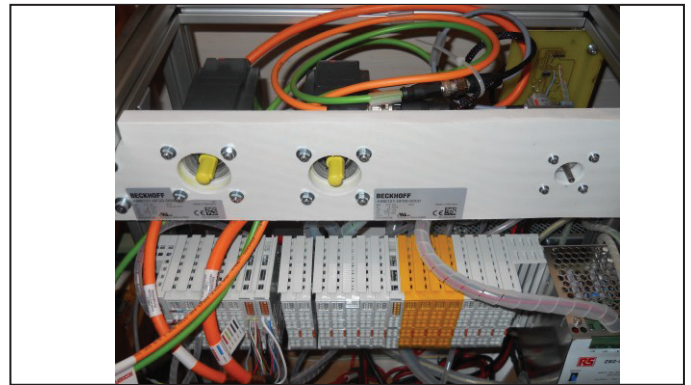


Fig. 15. Programming a safety configuration which includes an integrated safety module similar to figure 13 but from a different manufacturer.



Fig. 16. Programming a PSR and a classical PLC with technical communication between PLC and PSR. A Specific communication bus is used to connect the different safety devices to the PSR.

V. CONCLUSIONS AND PERSPECTIVES

The courses, as well as the practical works, have been developed according to the current needs of the industry and to prepare the students for the new issues related to the transition to the industry 4.0. The perspectives of this works are oriented to develop new pieces of training on safe robotics, safe motions (abrupt stop, safe reaction, safely limited speed, ...), redundancy (redundant ring) for high availability and cybersecurity. As a first step, future work will focus on safety motion control, these include integrated motion control functions like safe torque off, safe stop 1 and 2, safety limited speed and safe maximum speed by using PLCOpen safety axis modules with integrated safety functions. The different servo drives will communicate via Ethernet with safety on Ethernet.

REFERENCES

- [1] K.T. Kosmowski, "Functional safety concept for hazardous systems and new challenges", Journal of Loss Prevention in the Process Industries vol. 19, Issues 2-3, March-May 2006, pp. 298-305.
- [2] K.T. Kosmowski "Functional Safety Analysis including Human Factors", International Journal of Performability Engineering Vol. 7, No. 1, January, 2011, pp. 61-76.
- [3] H. Kagermann, W. Wahlster and J. Helbig "Recommendations for implementing the strategic initiative INDUSTRIE 4.0", National Academy of Science and Engineering, Apr. 2013.